Physics-Guided Generative Design for CDK2 Selectivity

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Potency ≠ Selectivity

Related kinases share similar ATP-binding sites, so a molecule predicted to bind CDK2 may also bind unwanted kinases.

Can ALGen preserve predicted CDK2 binding while avoiding related kinases?
Tested here through docking-guided molecular selection

METHOD

Which off-targets can we trust?

10 antitarget kinases
from measured CDK2 cross-activity

Receptor ensembles
1,005 structures screened

Reliable and useful?
docking-signal screening

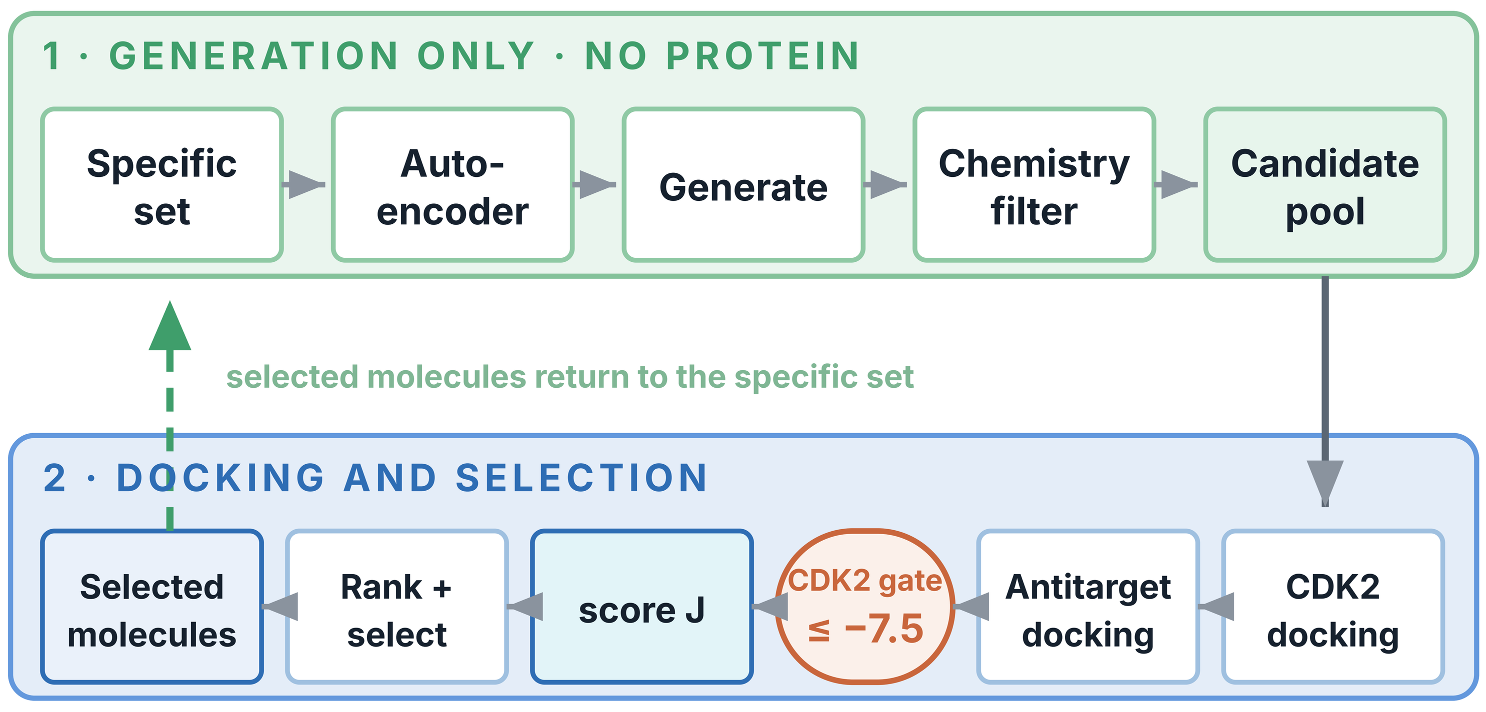
3 scored · 5 held out

RELIABLE → separates known binders
USEFUL → adds selectivity information

SCORED DYRK1A · GSK3B · PIM1
HELD OUT CDK1 · CDK5 · CDK6 · CDK7 · CDK9

scored enter J · held-out only evaluated

How does ALGen use them?



Docking guides selection, not gradients.
Protein information enters only through the molecules selected and recycled.

What does J reward?

J

+ TARGET
strong CDK2 binding

− ANTITARGET
binding to unwanted kinases

− PROMISCUITY
broad non-specific chemistry

+ SIZE CONTROL
prevents trivial shrinking/ enlargement

J = target − β_{AT}·antitarget − promiscuity + size

β_{AT} = 1 ON | β_{AT} = 0 OFF

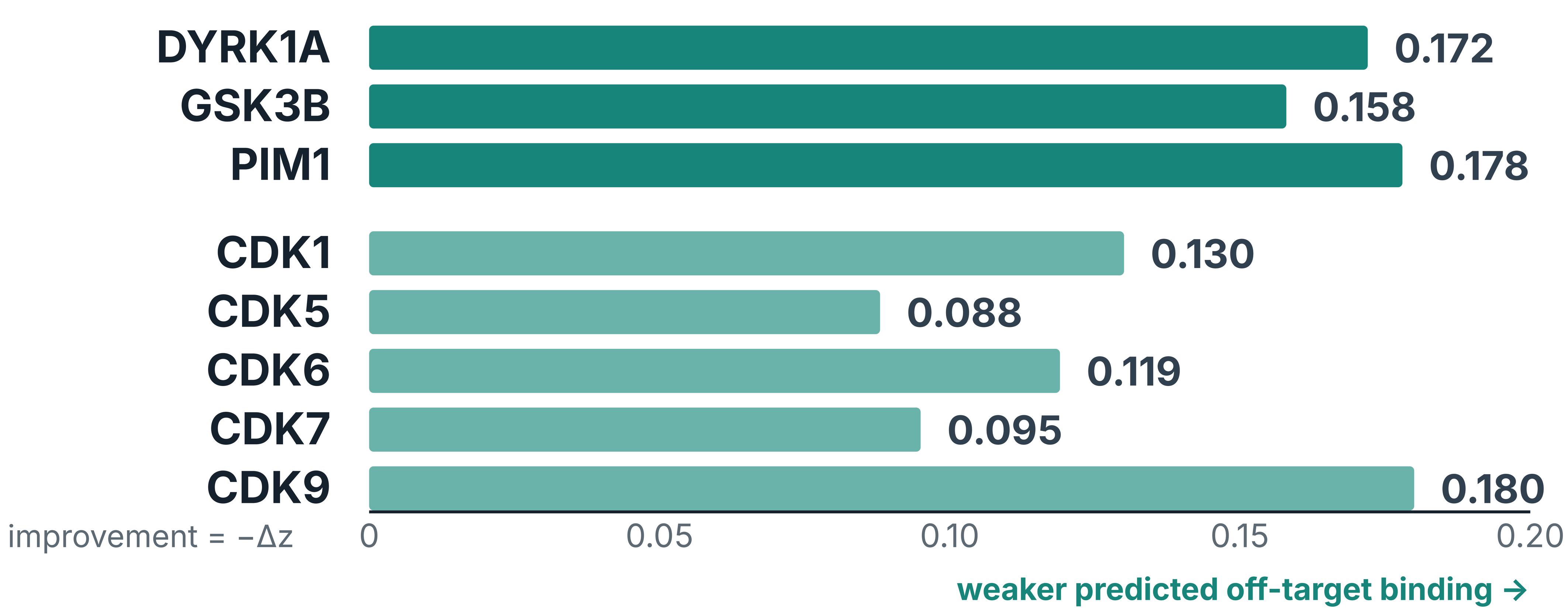
RESULT

SELECTION-LEVEL PROOF OF CONCEPT

Antitarget guidance changes molecular selection

Same candidate pool. Same selection budget. Only β_{AT} changes.

Off-target binding weakens across all 8 receptors

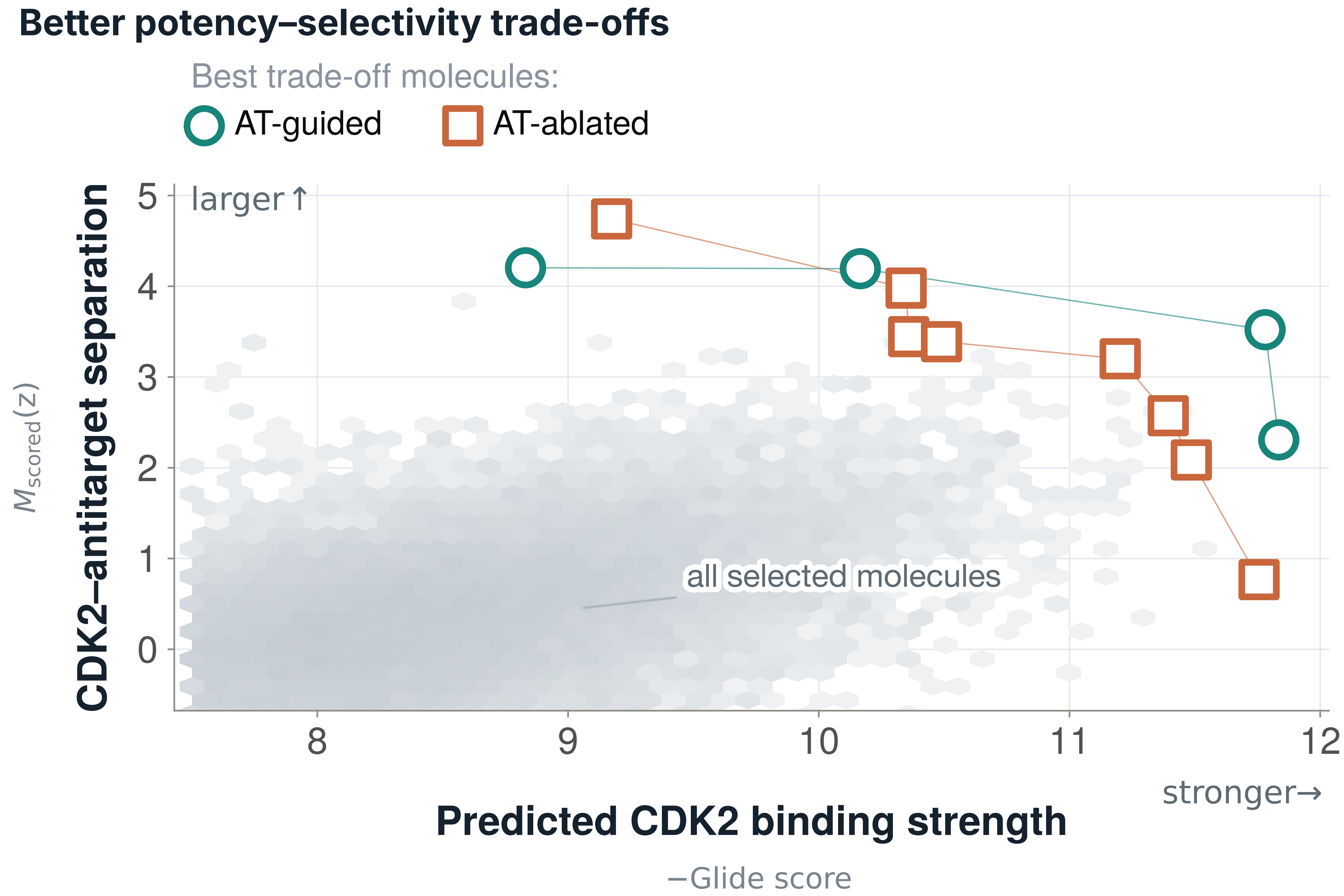


8 / 8
RECEPTORS IMPROVE
all 8 reported intervals exclude zero

13.95%
SELECTION CHANGED
279 of 2,000 from one 26,062-molecule pool

−0.058 z
CDK2 COST
slightly weaker predicted target binding

Better potency–selectivity trade-offs



Take-home

✓ **Selection shifts**
AT guidance changes which molecules survive.

✓ **8 / 8 off-targets improve**
with a small predicted CDK2 cost.

→ **Next: replicated trajectories**
test persistence through later generation.

−0.099 z observed ≈ ±0.102 z trajectory variability · 1 trajectory per condition

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predicted, docking-based selectivity · not measured biochemical selectivity
ALGen implementation · github.com/IFilella/ALGen-1

FULL PROJECT DETAILS
methods · figures · thesis · code